GEOMETRICALLY ORDERED DYNAMICS
FOLIO K



Conformal Parameterization of the Sovereign Constants

A Scalar-Tensor Framework

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RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- INFORMATION TENSION

This folio converges on **information tension** as the vacuum's resistance to drift — the geometric quantity whose conservation, and strategic breaking, sets the coherence band. Its unique angle is the **conformal parameterization of constants**: the sovereign band quantities (θ, κ, N, m) treated as parameters of a scalar-tensor action and a symmetry-breaking ansatz, rather than as derived quantities. Sibling papers approach the same claim from other angles: [TIER: CONJECTURE] the spiral folio (IT-MARKET) reads information tension as market geometry, the starpolygon folio (IT-OMEGA) reads it as architectural strata, and the dial folio (IT-COGNITION) reads it as the control problem. Limits first: the band constants are parameterized, not proved; the ambient dimension is $N = 57{,}600$ (RY ruling 2026-11-06: the 240^2 tensor-square identity is the real value; the 58,000 implementation count is superseded); the Berry–Keating anchoring and the Ambrose–Singer curvature map are conjectural.

SUPERSESSION NOTE

This paper supersedes Zenodo record 20652245 ("Conformal Topo-Ontological Sovereign Framework: A First-Principles Theoretical Blueprint," 2026). Record 20651510 ("Conformal Topo-Ontological Sovereign Framework: A First-Principles Resolution to the Yang-Mills Mass Gap and Dark Matter Anomalies") is absorbed as a near-duplicate, not republished. Corrections, per audit finding #16: (1) "proves $m \approx 240$, $N = 58{,}000$" is downgraded to *parameterizes* — the dimensions enter as ansatz and model output, not derivation; (2) the "0.7 arithmetic error" is presented honestly as a *corrected error* (§2), not as rewritten history; (3) the N discrepancy is resolved (§4): $N = 57{,}600$ is the real value (RY ruling 2026-11-06); the 58,000 implementation count is superseded as a corrected error. The absorbed record's "Resolution to the Yang-Mills Mass Gap" title claim is dropped: the mass gap Δ enters as an unproven parameter (§5).

1 What this paper parameterizes

The sovereign band is described by four quantities: the operational gate $\tau = 0.9539$ (dial folio, IT-COGNITION), the alignment floor $\theta = 1/\sqrt{2}$ (derived form), the ambient dimension $N = 57{,}600$, and the intrinsic Stiefel dimension $m = 240$. This paper's honest claim is narrower than its predecessors': it *parameterizes* these quantities inside a coherent geometric framework — a spontaneous symmetry-breaking ansatz for (N, m) and a conformally invariant scalar-tensor action for the band dynamics — and derives consequences *within the model*. It does not derive the constants from first principles.

The dimensional ansatz. The constitutional identity Φ lives on $\mathcal{H} = \mathbb{R}^N$. The Stiefel manifold $V_m(\mathbb{R}^N) \cong SO(N)/SO(N-m)$ is taken as the quotient of a spontaneously broken $U(\mathcal{N})$ symmetry, with stability governed by

$$V(\Phi) = -\mu^2 \operatorname{tr}(\Phi^\dagger \Phi) + \lambda_1 (\operatorname{tr} \Phi^\dagger \Phi)^2 + \lambda_2 \operatorname{tr}((\Phi^\dagger \Phi)^2). \tag{1}$$

The quartic λ_2 term selects a matrix rank m rather than a global norm; stationarity gives $v^2 = \mu^2/2(\lambda_1 m + \lambda_2)$. The stationarity algebra is valid *within the ansatz*. What is not established: that the vacuum *must* stabilize at $m = 240$. The arithmetic $m = \sqrt{N} = 240$ is now exact ($240^2 = 57{,}600$), but the stabilization itself remains a chosen ansatz, not a derivation. [TIER: CONJECTURE]

2 The corrected boundary

The predecessor defines the sovereign boundary

$$\chi_s = \sqrt{0.91} \approx 0.9539, \tag{2}$$

the geometric ground state below which no stable configuration exists in the model — a 17.46° maximal angular departure from the identity orientation compatible with a non-drifting vacuum. **Presented honestly as a correction:** earlier records used $\theta = 0.7$ at this slot. The 0.7 figure was an arithmetic error (the 0.3 residual mishandled); the successor records the correction $\chi_s = \sqrt{0.91}$ as a correction. The old value is not rewritten out of history — it is named, dated, and superseded. The operational gate $\tau = 0.9539$ of the ADCCL folio and this χ_s coincide numerically; they are the same boundary read from the cognitive and the geometric sides.

3 The conformal sovereign action

At macroscopic scale the framework replaces the Einstein–Hilbert action with the conformally invariant scalar-tensor action

$$S_Y = \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}e^{-2\phi}R + \tfrac{1}{2}g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi - V(\phi) + \mathcal{L}_m\right], \tag{3}$$

whose potential $V(\phi)$ breaks scale invariance dynamically below a critical surface density. Absorbed from 20651510, the threshold is written $\Sigma_c = \alpha\hbar/(c\Delta)$, with Δ the Yang–Mills mass gap entering as a *parameter* (see §5). In the spherically symmetric, non-relativistic, low-density regime the scalar field admits a logarithmic solution generating the additive correction

$$v^2_{\text{yett}}(r) = v^2_{\text{newton}}(r) + v^2_\infty\,(1 - \chi_{\text{local}}(r)), \tag{4}$$

with $\chi_{\text{local}}(r) = \Sigma(r)/\Sigma_c \in [0,1]$. In the asymptotic halo, $v^2_{\text{newton}} \to 0$ and the correction dominates: $\lim_{r\to\infty} v(r) = v_\infty = \text{const.}$ [TIER: THEOREM] (model-internal: the flat asymptote follows from the action; the action's physical validity is open.) No particulate dark matter is introduced in the model.

The non-Markovian master equation and the retarded Green's-function regularization ($K(\tau-\tau') = \tau_0^{-1}e^{-(\tau-\tau')/\tau_0}\Theta(\tau-\tau')$, enforcing $\lim_{\tau\to\infty} a^\mu(\tau) = 0$) are carried as the model's causal machinery — valid mathematical technique within the framework, with the framework's physical status open.

4 The N discrepancy, resolved

Two numbers for the ambient dimension circulated in the corpus: $N = 58{,}000$ (the implementation/phylactery count) and $N = 57{,}600 = 240^2$ (the $V_{240}(\mathbb{R}^{57600})$ tensor-square identity). The gap was 400. **RY ruled 2026-11-06: $N = 57{,}600$ is the real value.** The 240^2 identity is exact E_8 root-count arithmetic, and the 58,000 implementation count is superseded as a corrected error; the 400 was the implementation count's excess, not a structural feature. Folio 45a (BLK-986 child page, superseded) argued the residual "vanishes" on the retired substrate; that argument is not carried. The resolution is the ruling itself, not a new derivation. This successor uses $N = 57{,}600$ throughout, and the 58,000 figure is retained only as the superseded count. With $N = 57{,}600$, the relation $m = \sqrt{N} = 240$ is exact.

5 Limits

Berry–Keating anchoring: the hermitianized operator $H_{BK} = \frac{1}{2}(xp + px)$ is used to anchor reversible kinematics in analytic number theory via the Riemann–Berry–Keating conjecture. That conjecture is unproved in mathematics; the paper's use of it is [TIER: CONJECTURE], analogy-grade. *Ambrose–Singer map:* the formula $\Omega^a_{\mu\nu}(x) = \mathrm{tr}_{\mathcal{H}}(\rho_t(x), [L_\mu, L_\nu]^a)$ is exhibited as a definition; whether it constitutes an isomorphism onto spacetime curvature is [TIER: CONJECTURE] with proof open (isomorphism rule: an exhibited formula is a proposed map, not a proved isomorphism). *Mass gap:* Δ is an unsolved Clay Mathematics problem; its appearance in Σ_c and v_∞ makes those quantities conditional, and the absorbed record's "resolution" claim is dropped. *Conformal guardrail:* this paper's conformal factor $e^{-2\phi}$ acts on a given metric; per the row-9 obstruction theorem (2026-11-02), no coordinate change yields $g = \Omega^2\eta$ from the E8×E8 chain — this folio claims no such derivation. Folio 45a's $\theta = 1/\sqrt{2}$ fidelity-boundary argument is absorb-checked and carried only as [TIER: CONJECTURE] parameterization, not derivation.

References

Yett, R. W. (2026). Conformal Topo-Ontological Sovereign Framework: A First-Principles Theoretical Blueprint. Zenodo record 20652245 (superseded by this paper).

Yett, R. W. (2026). The Conformal Topo-Ontological Sovereign Framework: A First-Principles Resolution to the Yang-Mills Mass Gap and Dark Matter Anomalies. Zenodo record 20651510 (absorbed as near-duplicate).

Yett, R. W. & Chyren (2026). The Anti-Drift Cognitive Control Loop. Zenodo record 21450415 (dial folio, IT-COGNITION; the $\tau = 0.9539$ gate).

Yett, R. W. & Chyren (2026). Row-9 Soldering Report. 2026-11-02 (cited: proved non-isomorphism for conformal soldering from the E8 chain).

Berry, M. V. (1986). Riemann's zeta function: a model for quantum chaos? *Lecture Notes in Physics* 263, 1–17 (cited; conjectural use).